$a = 1.27 \approx 1.3\ ms^{-2}$

19 **Ans: B**

Because an electron is **negatively** charged, the **electric force pushes it in the opposite direction of the electric field lines.** Therefore, the electron is pushed **from a region of lower potential toward a region of higher potential.** Hence the electric potential it experiences **increases**.

When the electron is released from rest, the electric field does work on it, causing it to **accelerate**. As it accelerates, its **kinetic energy increases**. Because **total energy must be conserved,** an increase in kinetic energy must come **from a decrease in ~~stored~~ its potential energy.** Hence the potential energy **decreases**.